



EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Master of Science in Computer Science (Machine Learning)

Atlanta, GA

Aug. 2025 - Dec. 2026

Relevant Coursework: Machine Learning, Statistical Modeling, Database Systems, Natural Language Processing

SEOUL NATIONAL UNIVERSITY (SNU)

Bachelor of Science in Artificial Intelligence, Bachelor of Arts in Linguistics

Seoul, South Korea

Mar. 2019 - Feb. 2025

Relevant Coursework: AI, Computer Vision, Computational Linguistics, Data Structures, Algorithms, Databases, Probability Theory

EXPERIENCE

GT Human-Centered Artificial Intelligence Lab, *Research Intern* (Advisor: Prof. Mark Riedl)

Feb. 2026 - Present

- Studied training data influence on LLM behavior via causal intervention, developing machine unlearning pipelines to remove targeted data and measure downstream effects; second author on paper accepted to COLM 2026
- Constructed reproducible sampling pipelines over a stratified ~5B-token corpus, enabling scalable controlled experiments for LLM evaluation.

GE Vernova, *Data Scientist Intern*

Jun. 2026 - Aug. 2026

- Built and deployed production LLM services for enterprise workflows, designing evaluation pipelines over 1k+ criteria to measure relevance and instruction adherence (97.4%-98.2%)

Salesforce, *Data Scientist Intern*

Jan. 2025 - Jun. 2025

- Developed predictive scoring and market prioritization models across 50+ industries to guide medium-to-long-term sales strategy, directly informing headcount allocation and org structure decisions on which industries to target

SNU SKI-ML Lab, *Research Intern* (Advisor: Prof. Jay-yoon Lee)

Jun. 2024 - Jan. 2025

- Co-authored *LaSEr-Edit* on span-level text editing; built evaluation pipelines across LLaMA, Phi, Qwen to diagnose weaknesses and guide method refinement, improving fluency (+23% BERTScore) and reducing perplexity (-22%)
- Built synthetic logical-consistency datasets (rule-based, GPT-rendered) for fine-tuning; improved constraint satisfaction (+31%) via prompt engineering

SKILLS

NLP & LLM: LLM Fine-tuning, RAG, Machine Unlearning, Controlled Text Generation, Evaluation, Information Extraction

Retrieval & Ranking: Vector Search, ANN Search, Embedding-based Retrieval, Ranking & Recommendation, LangChain

Computer Vision & Multimodal: Object Detection (Faster R-CNN), OCR, Diffusion Models, Multimodal Fusion

Programming & Infra: Python (PyTorch, Scikit-learn, Hugging Face), SQL, Java, C++ | AWS, GCP, Databricks, Docker, Git

PROJECTS

Hybrid Vector Search & RAG System | Georgia Tech, CS 6400: Database Systems & Design

- Designed embedding-based retrieval system combining vector similarity search (FAISS, IVF-PQ ANN) with structured filtering (SQLite) on 150K+ documents, achieving 3-30× latency reduction through adaptive query strategies
- Engineered an ML-based ranking/routing layer (95%+ accuracy) selecting optimal retrieval strategy, integrated into a LangChain RAG pipeline for LLM question answering

Personalized Recommendation System via Ensemble Modeling | Hyundai Home Shopping Networks

- Developed a recommendation and ranking system combining LightGBM with SASRec (sequential recommender), boosting target-user purchase rate by 42%; engineered user/product embeddings and deployed on AWS

Korean Vehicle License Plate Recognition | SNU Computer Vision (2nd / 80 teams)

- Implemented an end-to-end vision system (Faster R-CNN + EasyOCR) in PyTorch for detection + OCR, achieving 96.5% character-level accuracy; curated a custom Korean license plate dataset to address data scarcity

Multimodal Behavioral Monitoring for Mental Health | Georgia Tech, CS 8803: Sonification Lab Research Project

- Built a multimodal representation system fusing vision (facial dynamics), audio (prosody), and text (STT semantics) signals for stress prediction, evaluating robustness under noisy real-world conditions

PUBLICATIONS

Same Persona, Different Language, Different Deal: Cross-Lingual Effects in LLM Negotiation (Under Review)

Where Does Social Reasoning Come From? Capability Provenance in Language Models, COLM 2026 [[arXiv](#)]

LaSEr-Edit: Localized Span-level Error Editing with Energy-based Localization, TMLR 2026 (Under Review) [[arXiv](#)]

Data-efficient End-to-end Information Extraction for Statistical Legal Analysis, EMNLP Workshop (NLLP) 2022 [[arXiv](#)]